

RESEARCH ARTICLE

Multivector (MV) Functions in Clifford Algebras of Arbitrary Dimension: Defective MV Case

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ABSTRACT

Explicit formulas to calculate MV functions in a basis-free representation are presented for an arbitrary Clifford geometric algebra $Cl_{p,q}$. The formulas are based on analysis of the roots of minimal MV polynomial and covers defective MVs, that is, the MVs that have non-diagonalizable matrix representations. The method may be generalized straightforwardly to matrix functions and to finite dimensional linear operators. The results can find wide application in Clifford algebra analysis.

1 | Introduction

Computation of matrix functions often arises as an important step in problems related to physical, economical, biological, and so forth processes. In geometric algebra (GA), this is equivalent to computation of function of a multivector (MV) [1–3]. A vast topic on matrix functions is well covered in the book [4].

If matrix is non-defective (diagonalizable), a typical computation step makes a start with a diagonalization procedure. Then, the function of diagonal matrix is straightforward to compute. In the non-diagonalizable case, the procedure requires Jordan decomposition, which introduces a lot of complications. In this article, we will focus on how to compute defective MV function in real $Cl_{p,q}$. The article presents an extension of our previous article [5] (which does not require neither explicit diagonalization nor Jordan decomposition procedure) to the case of defective MV or matrix. The described method can be used to verify our results of computation of MV functions in low dimensional ($n \leq 3$) Clifford algebras presented in [6, 7].

The matrix functions in general can be computed by a number of different ways [8–14]. Our approach is based on the method where renovation of spectral basis is employed and which belongs to the class of *polynomial methods* [8]. In particular, the latter method is based on the minimal polynomial of considered MV. Then, instead of generalized spectral decomposition procedure, which was used in [8] to find explicit basis expansion coefficients, we provide recursive formulas that greatly simplify the most problematic step in the computation. We believe that exact (closed form) formulas for exponentials and other functions for low dimensional cases investigated in [11, 12, 14], including matrices that are representations of some Lie groups or have some special symmetries [9, 10], now become relatively simple to compute.

In Section 2, the methods to generate characteristic polynomials in $Cl_{p,q}$ algebras characterized by arbitrary signature $\{p, q\}$ are discussed. The minimal polynomial is introduced in Section 3. In Section 4, we shortly remind classical method [8], and Section 5 presents main results of the article. Examples are

Abbreviations: GA, geometric (Clifford) algebra; MV, multivector; nD, n -dimensional vector space.

given in Sections 6–8 followed by Conclusion and Perspectives in Section 9.

2 | Notation and Characteristic Polynomial of MV

In GA, the geometric product of orthonormalized basis vectors $\mathbf{e}_i$ and $\mathbf{e}_j$ satisfy [2] the anti-commutation relation, $\mathbf{e}_i \mathbf{e}_j + \mathbf{e}_j \mathbf{e}_i = \pm 2\delta_{ij}$, where space denotes usual geometric product. For a mixed signature $Cl_{p,q}$ algebra, the squares of basis vectors are $\mathbf{e}_i^2 = +1$ for first p vectors, $i = 1, 2, \dots, p$, and $\mathbf{e}_j^2 = -1$ for remaining q vectors, $j = p+1, p+2, \dots, p+q$. The sum $n = p+q$ is equal to dimension of the vector space. The general MV is expressed as

$$\mathbf{A} = a_0 + \sum_i a_i \mathbf{e}_i + \sum_{i < j} a_{ij} \mathbf{e}_{ij} + \dots + a_{1\dots n} \mathbf{e}_{1\dots n} = \sum_{J=0}^{2^n-1} a_J \mathbf{e}_J, \quad (1)$$

where $a_i, a_{ij}, \dots$ are the real coefficients. The number of subscripts in the basis element $\mathbf{e}_{\dots}$ indicates the grade of the element. The ordered set of indices denoted by a single capital letter J is referred to as a multi-index. The basis elements $\mathbf{e}_{ij}, \dots$ are always assumed to be listed in the inverse degree lexicographic order; that is, the element with a lower grade is listed before all elements of higher grades while lexicographically once the grade of both elements is the same. For example, when $p+q=3$, the basis elements are listed in the following order $\{1, \mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \mathbf{e}_{12} \equiv \mathbf{e}_1 \mathbf{e}_2, \mathbf{e}_{13} \equiv \mathbf{e}_1 \mathbf{e}_3, \mathbf{e}_{23} \equiv \mathbf{e}_2 \mathbf{e}_3, \mathbf{e}_{123} \equiv \mathbf{e}_1 \mathbf{e}_2 \mathbf{e}_3 \equiv I\}$; that is, both the number of indices and their values always increase from left to right.

Characteristic polynomial $\chi_A(\lambda)$ of MV $\mathbf{A}$ of variable λ plays an important role in many applications. We will mainly use it for general considerations and for comparison with minimal polynomial of MV, which is described in next section. Every MV $\mathbf{A} \in Cl_{p,q}$ has a characteristic polynomial $\chi_A(\lambda)$ of a degree d in $\mathbb{R}$,

$$\chi_A(\lambda) = \sum_{k=0}^d C_{(d-k)}(\mathbf{A}) \lambda^k. \quad (2)$$

where $d = 2^{\lfloor \frac{n}{2} \rfloor}$ is the integer, $n = p+q$. In particular, $d = 2^{n/2}$ if n is even, and $d = 2^{(n+1)/2}$ if n is odd.

We will take¹ $C_{(0)}(\mathbf{A}) = -1$, then $C_{(1)}(\mathbf{A})$ will represent MV's trace: $C_{(1)}(\mathbf{A}) = \text{Tr}(\mathbf{A}) = d \langle \mathbf{A} \rangle_0$, where $\langle \mathbf{A} \rangle_0$ is the scalar part of MV in (1), that is, $\langle \mathbf{A} \rangle_0 = a_0$. The largest coefficient $C_{(d)}(\mathbf{A})$ gives MV determinant with opposite sign, $C_{(d)}(\mathbf{A}) = -\text{Det}(\mathbf{A})$.

The characteristic polynomial can be computed in a number of ways. We will use recursive Faddeev–LeVerrier–Souriau algorithm [15–17], where each recursion step produces one of the coefficients $C_{(k)}(\mathbf{A})$ of the polynomial (2). The first recursion gives $C_{(1)}(\mathbf{A})$. Each subsequent step produces the coefficient $C_{(k)}(\mathbf{A}) = \frac{d}{k} \langle \mathbf{A}_{(k)} \rangle_0$ and a new MV $\mathbf{A}_{(k+1)} = \mathbf{A}(\mathbf{A}_{(k)} - C_{(k)}(\mathbf{A}))$ according to

$$\begin{aligned} \mathbf{A}_{(1)} &= \mathbf{A} \rightarrow C_{(1)}(\mathbf{A}) = \frac{d}{1} \langle \mathbf{A}_{(1)} \rangle_0, \\ \mathbf{A}_{(2)} &= \mathbf{A}(\mathbf{A}_{(1)} - C_{(1)}(\mathbf{A})) \rightarrow C_{(2)}(\mathbf{A}) = \frac{d}{2} \langle \mathbf{A}_{(2)} \rangle_0, \\ &\vdots \\ \mathbf{A}_{(d)} &= \mathbf{A}(\mathbf{A}_{(d-1)} - C_{(d-1)}(\mathbf{A})) \rightarrow C_{(d)}(\mathbf{A}) = \frac{d}{d} \langle \mathbf{A}_{(d)} \rangle_0. \end{aligned} \quad (3)$$

The last step, as mentioned, returns the determinant with opposite sign: $-\text{Det}(\mathbf{A}) = \mathbf{A}_{(d)} = C_{(d)}(\mathbf{A}) = \mathbf{A}(\mathbf{A}_{(d-1)} - C_{(d-1)}(\mathbf{A}))$. At $(d+1)$ step one gets the identity, $\mathbf{A}_{(d+1)} = \mathbf{A}(\mathbf{A}_{(d)} - C_{(d)}(\mathbf{A})) = 0$, known as Cayley–Hamilton theorem. In particular, it states that if we replace polynomial variable λ by multivector $\mathbf{A}$ in the characteristic polynomial (2), we still get zero:

$$\sum_{k=0}^d \mathbf{A}^k C_{(d-k)}(\mathbf{A}) = \mathbf{A}^d C_{(0)}(\mathbf{A}) + \mathbf{A}^{d-1} C_{(1)}(\mathbf{A}) + \dots + C_{(d)}(\mathbf{A}) = 0. \quad (4)$$

The above identity allows to write down multivector power $\mathbf{A}^d$ (and, of course, all higher powers) as a linear combination of multivectors $\mathbf{A}^0, \mathbf{A}, \mathbf{A}^2, \dots, \mathbf{A}^{d-1}$, which correspond to respective matrices.²

The Cayley–Hamilton theorem also can be interpreted as a statement about linear dependency of vectors in higher dimensional space. Indeed, since geometric multiplication of MVs gives another MV of the same algebra, we can write all coefficients at (sorted) basis elements of MV in a list and interpret them as the components of some vector in 2^n dimensional linear space. Alternatively, since $m \times m$ matrix multiplication also yields a matrix of same dimension $m \times m$ we can reorder (flatten) $m \times m$ elements of a matrix into a column/row of length m^2 and also interpret them as vectors in some other linear space. Then Cayley–Hamilton theorem simply states that vectors constructed in both cases become linearly dependent if their number exceeds value d . We see, that in vector interpretation the role Cayley–Hamilton theorem plays (by setting maximal value d) is very similar to a role that vector space dimension n plays by restricting maximal number of independent vectors in nD. Therefore, we can use the same universal algorithm to search for vector dependency. This will help us to understand the method of computation of the minimal polynomial of MV in next section, which is just a variant of search of independent vectors in given vector space.

Let's now consider a Taylor series expansion³ of some MV function $f(\mathbf{A})$ around a regular point of the same function $f(x)$ of scalar argument, for example, around zero

$$f(\mathbf{A}) = f(0) + f'(0)\mathbf{A} + \frac{1}{2!} f''(0)\mathbf{A}^2 + \dots + \frac{1}{k!} f^{(k)}(0)\mathbf{A}^k + \dots \quad (5)$$

The Cayley–Hamilton theorem ensures that all powers equal or large than $\mathbf{A}^d$ can be expressed in terms of lower ones:

$$\begin{aligned} f(\mathbf{A}) &= (f(0) + \dots) + (f'(0) + \dots)\mathbf{A} + \left(\frac{1}{2!} f''(0) + \dots \right) \mathbf{A}^2 \\ &\quad + \dots + \left(\frac{1}{(d-1)!} f^{(d-1)}(0) + \dots \right) \mathbf{A}^{d-1}, \end{aligned} \quad (6)$$

where the coefficients of linear combination of all higher powers of MV now have been moved inside lower expansion coefficients $\left(\frac{1}{k!} f^{(k)}(0) + \dots \right)$ at $\mathbf{A}^k, k \leq (d-1)$. The sum, nevertheless, may be infinite. The question, therefore, is “can such a sum be summed up?” It appears that one can find a basis in which the sums at powers $\mathbf{A}^k, k \leq (d-1)$ become finite; that is, we can explicitly do the summation. The basis will be called a generalized spectral basis [8].

TABLE 1 | Minimal and characteristic polynomial factors in case of root λ_i of multiplicity k for trivial (left) and non-trivial (right) Jordan blocks of dimension k .

Jordan block	$\left. \begin{array}{cccc} \lambda_i & 0 & \cdots & 0 \\ 0 & \lambda_i & \cdots & 0 \\ & & \ddots & \\ 0 & \cdots & & \lambda_i \end{array} \right\}^k$	$\left. \begin{array}{cccc} \lambda_i & 1 & 0 & \cdots & 0 \\ 0 & \lambda_i & 1 & \cdots & 0 \\ & & & \ddots & \\ 0 & \cdots & & \lambda_i & 1 \\ 0 & \cdots & 0 & & \lambda_i \end{array} \right\}^k$
$\mu_{\hat{A}(x)}$	$(x - \lambda_i)$	$(x - \lambda_i)^k$
$\chi_{\hat{A}(x)}$	$(x - \lambda_i)^k$	$(x - \lambda_i)^k$

3 | Minimal Polynomial of MV

In the matrix theory, a most important polynomial is the so-called minimal polynomial, which we will denote $\mu_{\hat{A}}(\lambda)$. It establishes conditions for diagonalizability of matrix $\hat{A}$. Matrices will be denoted by hats in order to distinguish them from MVs, which lack hats. The polynomial $\mu_{\hat{A}}(\lambda)$ may be defined for MV as well. It is well-known that matrix is diagonalizable (aka non-defective) if and only if the minimal polynomial of the matrix does not have multiple (repeated) roots, that is, when the minimal polynomial consists of product of distinct linear factors. We can use this condition as a definition and the criterion for diagonalizability (non-defectiveness) of MV too. Table 1 compares characteristic and minimal polynomial factors for trivial and non-trivial Jordan blocks in case of matrices $\hat{A}$.

In calculation of MV function $f(\mathbf{A})$, a central role is played by MV minimal polynomial $\mu_{\mathbf{A}}(\lambda)$. For MV $\mathbf{A} \in Cl_{p,q}$, it is a polynomial in variable λ of degree $d \leq 2^{\lceil \frac{p}{2} \rceil}$,

$$\mu_{\mathbf{A}}(\lambda) = \mu(\lambda) = \sum_{k=0}^d C_{(d-k)}(\mathbf{A}) \lambda^k, \quad (7)$$

where the coefficients⁴ $C_{(d-k)}(\mathbf{A})$ characterize MV/matrix, or a general linear operator. We will take the coefficient at λ^d to be normalized to unity, that is, $C_{(0)}(\mathbf{A}) = 1$ (for characteristic polynomial we used $C_{(0)}(\mathbf{A}) = -1$). It is well known [18] that the minimal polynomial $\mu_{\mathbf{A}}(\lambda)$ divides the characteristic polynomial $\chi_{\mathbf{A}}(\lambda)$; therefore, in general, the former is of the same or lower degree than later (Table 1). If all roots of characteristic equation $\chi_{\mathbf{A}}(\lambda) = 0$ are different, in other words, the roots have multiplicity 1, then $\chi_{\mathbf{A}}(\lambda)$ and $\mu_{\mathbf{A}}(\lambda)$ may differ by constant factor (normalization coefficient) only.

To have shorter notation, instead of $\mu_{\mathbf{A}}(\lambda)$, we shall write μ , since μ depends on MV $\mathbf{A}$ indirectly through coefficients C_k . We will also introduce an alternative notation with superscript, $\mu(\lambda) = \mu^{(0)}(\lambda)$, in order to make the notation compatible with that for k -th derivative of minimal polynomial, $\mu^{(k)}(\lambda) = \frac{1}{k!} \frac{d^k \mu(\lambda)}{d\lambda^k}$. Taking into account that the factorial of zero is $0! = 1$, the notation $\mu^{(0)}(\lambda)$ simply means a “zero” derivative of minimal polynomial, that is, $\mu^{(0)}(\lambda) \equiv \mu(\lambda)$. The first derivative of $\mu(\lambda)$ then is $\mu^{(1)}(\lambda) = \frac{1}{1!} \frac{d\mu(\lambda)}{d\lambda} = \sum_{k=1}^d k C_{(d-k)}(\mathbf{A}) \lambda^{k-1} = \sum_{k=0}^{d-1} (k+1) C_{(d-k-1)}(\mathbf{A}) \lambda^k$, where in the last sum the summation index was shifted to start from zero in order to be consist with the formula (16) in paper [5] (see factor β therein).

Algorithm 1 explains in terms of GA how the MV minimal polynomial can be programmed.⁵

Algorithm 1 Minimal polynomial of MV.

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1: procedure MINIMALPOLY( $\mathbf{A}, x$ ) ▷ Input is the MV
    $\mathbf{A} = \sum_{j=0}^{2^n-1} a_j \mathbf{e}_j$  and polynomial variable  $x$ 
2:   nullSpace={}; lastProduct=1; vectorList={}; ▷
   Initialization; {} is an empty list
3:   while nullSpace=={} do ▷ Keep adding MV coefficient
     vectors to vectorList until the null space gets to be nonempty
4:     lastProduct ←  $\mathbf{A} \circ \text{lastProduct}$ 
5:     vectorList ← AppendTo[vectorList, toCoeffi-
     cientList[lastProduct]]
6:     nullSpace ← NullSpace[Transpose[vectorList]];
7:   end while ▷ Construct minimal polynomial from
     nullSpace coefficients and powers of input variable  $x$ 
8:
9:   if Length[nullSpace] ≤ deg( $\chi$ ) then ▷ If length exceeds
     the degree of characteristic polynomial, reduce degree by one
10:    return First[nullSpace] ·  $\{x^1, x^2, \dots, x^{\text{Length[nullSpace]}}\}$ 
11:  else
12:    return First[nullSpace] ·  $\{x^0, x^1, x^2, \dots, x^{\text{Length[nullSpace]}-1}\}$ 
13:  end if
14: end procedure

```

The algorithm was explained earlier in paper [5]; else, the computation closely follows the algorithm for a matrix minimal polynomial in [19]. Because, in GA, the orthonormal basis elements $\mathbf{e}_j$ are linearly independent by definition, we can construct vectors from MV coefficients, in an exactly the same way as it was described in the interpretation of Cayley–Hamilton formula (4). Algorithm 1 performs a search until the vectors constructed from coefficients at powers of MV become linearly dependent. Once this has been achieved,⁶ the **NullSpace[]** returns a list of coefficients of the linear combination. These coefficients then are multiplied by corresponding power of $\mathbf{A}$ and summed up to form a minimal MV polynomial.

Since output of *Mathematica* command **NullSpace[]** is a bit specific, it deserves more detailed explanation. Suppose we want to compute the minimal polynomial of the idempotent $\mathbf{a} = \frac{1}{2}(1 + \mathbf{e}_1)$ of $Cl_{3,0}$ algebra with orthonormal basis listed in Section 2. Following Algorithm 1, we start by forming list of components of $\mathbf{a}, \mathbf{a}^2, \mathbf{a}^3, \dots$. A first vector $\mathbf{a}$ added to nullSpace variable has components $\{1/2, 1/2, 0, 0, 0, 0, 0\}$. In the second step, we add vector components $\{1/2, 1/2, 0, 0, 0, 0, 0\}$ of $\mathbf{a}^2$. Since $\mathbf{a}$ is an idempotent, that is, $\mathbf{a}^2 = \mathbf{a}$, the components of $\mathbf{a}^2$ are exactly the same. The variable nullSpace now contains two identical vectors $\text{nullSpace} = \{\{1/2, 1/2, 0, 0, 0, 0, 0\}, \{1/2, 1/2, 0, 0, 0, 0, 0\}\}$,

which are, obviously, linearly dependent. The `NullSpace[]` returns $\{-1, 1\}$, which simply states that $-a + a^2 = 0$. The computed minimal polynomial of a , therefore, is $-x + x^2$, which coincides with the minimal polynomial of matrix representation of a by complex 2×2 Pauli like matrices $\hat{e}_1 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, $\hat{e}_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\hat{e}_3 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$. It should be noted, however, that this is not always the case. Indeed, since real $Cl_{3,0}$ Clifford algebra is represented by complex 2×2 matrices, one can easily find an example when the minimal polynomial of MV and the minimal polynomial of MVs matrix representation are distinct. For example, if we compute a minimal polynomial of an algebraic spinor $\psi = \frac{1}{2}c_1(\mathbf{e}_1 + 1) + \frac{1}{2}c_3(\mathbf{e}_2 - \mathbf{e}_{12}) + \frac{1}{2}c_2(\mathbf{e}_{123} + \mathbf{e}_{23}) + \frac{1}{2}c_4(\mathbf{e}_{13} - \mathbf{e}_3)$, which was computed using the idempotent a above, we get $\mu_\psi(x) = x^3 - 2c_1x^2 + (c_1^2 + c_2^2)x$. Obviously, it cannot be a minimal polynomial of the matrix representation, which is 2×2 complex matrix (and, therefore, cannot be of degree 3). Indeed, the matrix representation of the above spinor is $\hat{\psi} = \begin{pmatrix} c_1 + ic_2 & 0 \\ c_3 - ic_4 & 0 \end{pmatrix}$. The minimal polynomial $\mu_{\hat{\psi}}(x) = x^2 + (-c_1 - ic_2)x$ of this matrix is of degree 2. Note, that in contrast to real polynomial $\mu_\psi(x)$ of MV, the polynomial $\mu_{\hat{\psi}}(x)$ of the matrix representation has complex coefficients and is not suited for real Clifford algebra $Cl_{3,0}$, since we know that geometric multiplication of MVs with real coefficients can produce real expressions only. The same remark applies for characteristic polynomial in GA, which is now well understood [17].

The described method of computation of minimal polynomial must bring in a clear difference between the characteristic and minimal polynomials: The characteristic polynomial is able to provide general condition when generic vectors must become linearly dependent only, whereas the minimal polynomial finds strict linear relation between particular vectors. For example, while randomly generated 3 vectors (with integer coefficients) in 3D space are linearly independent in most cases, it may happen that in rare cases they all lie in the same 2D plane, or even be collinear. The characteristic polynomial cannot distinguish between these cases, while minimal polynomial sensitively represents each case.

At the end, we would like to comment briefly on computation of minimal polynomial in *Mathematica* language syntax (also refer to Algorithm 1). The full implementation is provided in GA package [20]. The command-functions that begin with a capital letter, `AppendTo[]`, `NullSpace[]`, `Transpose[]`, `Last[]`, `First[]`, and `Dot[]` (short form of which is `.`), are internal *Mathematica* commands. Exceptions are the symbol $\circ$ (geometric product) and `toCoefficientList[]` transformation. The latter is very simple. It takes a multivector A and constructs a vector (list) from its coefficients: `toCoefficientList` $[a_0 + a_1\mathbf{e}_1 + a_2\mathbf{e}_2 + \dots + a_I I] \rightarrow \{a_0, a_1, a_2, \dots, a_I\}$. A real job is done by *Mathematica* function `NullSpace[]`, which searches for linear dependency of the incrementally augmented vector list `vectorList` (see Algorithm 1). After such a list of vectors has been found (this is guaranteed by Cayley–Hamilton theorem) the function outputs the linear combination for which the sum of vectors turns to zero. The command `AppendTo` $[\text{vectorList}, \text{newVector}]$ appends the `newVector` to the list `vectorList` of already checked vectors.

4 | Calculation of MV Function by Classical Method

To help the reader grasp the main result presented in the next section, here, we will review the classical method how to find a generalized spectral basis and then apply it to compute the function of multivector, matrix, or operator [8].

Construction of the basis starts from minimal polynomial, which in a factorized form can be written as

$$\mu(x) = \prod_{i=1}^r (x - \lambda_i)^{m_i}. \quad (8)$$

Here, r stands for different eigenvalues λ_i of multiplicity m_i . The degree of the minimal polynomial $\mu(x)$ is $\sum_{i=1}^r m_i \leq 2^{\lfloor \frac{n}{2} \rfloor}$. For convenience, the roots are listed in increasing multiplicity order: $1 \leq m_1 \leq m_2 \leq \dots \leq m_r$. So, the last root has the largest multiplicity. The generalized spectral decomposition of the MV begins by decomposing the inverse of a minimal polynomial into partial fractions:

$$\begin{aligned} \frac{1}{\mu(x)} &= \sum_{i=1}^r \left(\frac{a_0}{(x - \lambda_i)^{m_i}} + \frac{a_1}{(x - \lambda_i)^{m_i-1}} + \dots + \frac{a_{m_i-1}}{(x - \lambda_i)} \right) \\ &= \frac{h_1(x)}{(x - \lambda_1)^{m_1}} + \dots + \frac{h_r(x)}{(x - \lambda_r)^{m_r}} \\ &= \frac{h_1(x)\psi_1(x) + \dots + h_r(x)\psi_r(x)}{\mu(x)}, \end{aligned} \quad (9)$$

where $h_i(x) = \sum_{s=0}^{m_i-1} a_s(x - \lambda_i)^s$ and $\psi_i(x) = \prod_{j \neq i}^r (x - \lambda_j)^{m_j}$; that is, we have divided each term of the sum into part $h_i(x)$ that includes the root λ_i and the part $\psi_i(x)$ that is free of the root λ_i . After comparison with the initial expression, we see that $\sum_{i=1}^r h_i(x)\psi_i(x) = h_1(x)\psi_1(x) + \dots + h_r(x)\psi_r(x) = 1$. Therefore, we can define a polynomials $p_i(t) = h_i(t)\psi_i(t)$ having the property $\sum_{i=1}^r p_i = 1$. Since $\psi_i(t)\psi_j(t)$ is divisible⁷ by $\mu(t)$ for all $i \neq j$, we also have $p_i p_j = 0$. Thus, we conclude that $p_i^2 = (p_1 + \dots + p_r)p_i = 1 p_i = p_i$. The obtained properties imply that the operators p_i make a set of mutually annihilating idempotents that realize partition of the unity.

Now, for each $1 \leq i \leq r$, let us define a new polynomial $q_i = (x - \lambda_i)p_i$. For non-repeating root $m_i = 1$, we see that $q_i = 0$. For repeated root of multiplicity $m_i > 1$, we have $q_i^{m_i-1} \neq 0$, but $q_i^{m_i} = 0$. Thus, a list $q_1, q_2, \dots, q_r$ is a set of nilpotents with corresponding nilpotency indices $m_1, m_2, \dots, m_r$. If $i \neq j$, we have $p_i q_j = 0$. Therefore, $p_i q_i = (p_1 + \dots + p_r)q_i = 1 q_i = q_i$, and one says that nilpotents $q_1, q_2, \dots, q_r$ are projectively related to idempotents $p_1, p_2, \dots, p_r$. As a result, we have got the following generalized spectral basis:

$$\left\{ \underbrace{p_1, q_1, q_1^2, \dots, q_1^{m_1-1}}_{m_1}, \dots, \underbrace{p_i, q_i, q_i^2, \dots, q_i^{m_i-1}}_{m_i}, \dots, \underbrace{p_r, q_r, q_r^2, \dots, q_r^{m_r-1}}_{m_r} \right\}, \quad (10)$$

where $p_i = p_i(x)$, $q_i = q_i(x)$ are polynomials. If the multiplicity m_i of root λ_i is 1, the corresponding basis blocks (indicated by underbraces) consist of a single polynomial p_i .

Lemma 1. The generalized spectral decomposition of linear operator can be written as

$$\mathbf{A} = \sum_{i=1}^r (\lambda_i + q_i) p_i. \quad (11)$$

We need to show that $\mathbf{A}p_i = (\lambda_i + q_i)p_i$. Since $q_i = (\mathbf{A} - \lambda_i)p_i$, $\mathbf{A}p_i = q_i + \lambda_i p_i = q_i p_i + \lambda_i p_i$, where we used the property $p_i q_i = q_i$. Because p_i is the idempotent, $p_i^2 = p_i$, after expansion, we have $\mathbf{A} \sum_{i=1}^r p_i = \mathbf{A}$.

Now, higher powers of $\mathbf{A}$ can be easily computed:

$$\begin{aligned} 1 &= p_1 + p_2 + \cdots + p_r \\ \mathbf{A} &= (\lambda_1 + q_1)p_1 + (\lambda_2 + q_2)p_2 + \cdots + (\lambda_r + q_r)p_r \\ \mathbf{A}^2 &= (\lambda_1 + q_1)^2 p_1 + (\lambda_2 + q_2)^2 p_2 + \cdots + (\lambda_r + q_r)^2 p_r \\ &\vdots \\ \mathbf{A}^{m-1} &= (\lambda_1 + q_1)^{m-1} p_1 + (\lambda_2 + q_2)^{m-1} p_2 + \cdots + (\lambda_r + q_r)^{m-1} p_r. \end{aligned} \quad (12)$$

To find $(\lambda_j + q_j)^k$, one can use the binomial theorem and property $q_i^{m_i} = 0$:

$$(\lambda_j + q_j)^k = \lambda_j^k + \binom{k}{1} \lambda_j^{k-1} q_j + \cdots + \binom{k}{m_j-1} \lambda_j^{k-(m_j-1)} q_j^{m_j-1}. \quad (13)$$

Here, $\binom{k}{p} = \frac{k!}{p!(k-p)!}$ denotes the usual binomial coefficient for $p \leq k$ and zero otherwise.

Once we know how to compute the powers of MV, we can apply this property to function series in MV argument (we suppose the function is analytic). For simplicity, let us assume that the function of a scalar (complex) argument, $g(z)$, has a convergent Taylor series in the neighborhood of the eigenvalue λ_i . Then, MV function $g(\mathbf{A})$ can be defined as Taylor series with scalar variable z replaced by MV⁸ $\mathbf{A}$; that is, $g(\mathbf{A}) = \sum_{k=0}^{\infty} \frac{g^{(k)}(\lambda_i)}{k!} \mathbf{A}^k = a_0 + a_1 \mathbf{A} + a_2 \mathbf{A}^2 + \cdots$. Using (12) and properties $p_i^2 = p_i$, $q_i^{m_i} = 0$, $p_i q_i = q_i$, $p_j q_i = 0$, we find:

$$\begin{aligned} g(\mathbf{A}) &= a_0(p_1 + p_2 + \cdots + p_r) \\ &+ a_1((\lambda_1 + q_1)p_1 + (\lambda_2 + q_2)p_2 + \cdots + (\lambda_r + q_r)p_r) \\ &+ a_2((\lambda_1 + q_1)^2 p_1 + (\lambda_2 + q_2)^2 p_2 + \cdots + (\lambda_r + q_r)^2 p_r) \\ &+ a_3((\lambda_1 + q_1)^3 p_1 + (\lambda_2 + q_2)^3 p_2 + \cdots + (\lambda_r + q_r)^3 p_r) + \cdots \\ &= (a_0 + a_1 \lambda_1 + a_2 \lambda_1^2 + a_3 \lambda_1^3 + \cdots) \\ &+ \left(a_1 + \binom{2}{1} a_2 \lambda_1 + \binom{3}{1} a_3 \lambda_1^2 + \cdots \right) q_1 \\ &+ \left(a_2 + \binom{3}{2} a_3 \lambda_1 + \binom{4}{2} a_4 \lambda_1^2 + \cdots \right) q_1^2 + \cdots \\ &+ \text{sums for rest } r-1 \text{ roots} \\ &= g(\lambda_1)p_1 + g'(\lambda_1)q_1 + \frac{1}{2!} g''(\lambda_1)q_1^2 + \cdots \\ &+ \frac{1}{(m_1-1)!} g^{(m_1-1)}(\lambda_1)q_1^{(m_1-1)} + \cdots \\ &+ g(\lambda_r)p_r + g'(\lambda_r)q_r + \frac{1}{2!} g''(\lambda_r)q_r^2 + \cdots \\ &+ \frac{1}{(m_1-1)!} g^{(m_1-1)}(\lambda_r)q_r^{(m_1-1)}, \end{aligned} \quad (14)$$

where the primes denote derivatives, $g'(\lambda_1) \equiv g^{(1)}(\lambda_1) = \frac{dg(z)}{dz}|_{z=\lambda_1}$ and so forth. Introduction of notation

$$\begin{aligned} g(\lambda_i + q_i) &= g(\lambda_i)p_i + g'(\lambda_i)q_i + \frac{1}{2!} g''(\lambda_i)q_i^2 \\ &+ \cdots + \frac{1}{(m_i-1)!} g^{(m_i-1)}(\lambda_i)q_i^{(m_i-1)}, \end{aligned} \quad (15)$$

then allows compactly write down the above finite sum as

$$g(\mathbf{A}) = \sum_{i=1}^r g(\lambda_i + q_i) p_i. \quad (16)$$

The classical formula (16) may be used to compute analytical functions of MV, matrices, or general linear operators. Note that almost all steps of symbolic manipulations (the decomposition of inverse of minimal polynomial, the Taylor expansion) can be done using formal scalar argument. The essential and most tricky part of these computations is, of course, the decomposition of inverse of minimal polynomial $1/\mu(x)$, which makes a core of all computations. The above description (intentionally) lacks any details on this subtle point. The only important feature of classical computation method we want to stress is the order in which polynomials p_i , q_i^k are computed. In particular, the polynomials p_i are computed at first. Only then it is possible to compute powers of q_i , q_i^2 , ... in increasing order. Because of latter property, the classical method requires operation of division (reduction) of polynomials. This is a difficult task when minimal polynomial happens to be an irreducible polynomial of high degree. In the article [8], the decomposition is implemented by using *Mathematica*'s universal command **Series[]** and the fact that the expansion series must terminate after m_i terms (the required program code is listed in the appendix of [8]). However, the **Series[]** command uses general state-of-the-art algorithms, where a short note on implementation of the command⁹ states that it “works by recursively composing series expansions of functions with series expansions of their arguments.” In the next section, we will show how to realize this decomposition explicitly, without any reference to command **Series[]**.

5 | MV Function in $Cl_{p,q}$ Algebra

This section presents main result of the article and extends our earlier efforts [5] to the case of non-diagonalizable MVs in a basis-free form.

Proposition 1. (MV function in a basis-free form). The function $f(\mathbf{A})$ of multivector argument $\mathbf{A} = \sum_{j=0}^{2^n-1} a_j \mathbf{e}_j$ in $Cl_{p,q}$ algebra is the MV that can be computed by the following explicit formula:

$$f(\mathbf{A}) = \sum_{i=1}^r \sum_{t=0}^{m_i-1} \left(\frac{d^t f(\lambda_i + \tau)}{d\tau^t} \Big|_{\tau=0} \right) Q_i^{m_i-1-t}(\mathbf{A}, \lambda_i), \quad (17)$$

where m_i denotes multiplicity of root λ_i (with the index i) of the minimal polynomial $\mu(\lambda)$ of the MV $\mathbf{A}$. The polynomial $\mu(\lambda)$ has r different roots. The symbol $\frac{d^t f(\lambda_i + \tau)}{d\tau^t} \Big|_{\tau=0}$ denotes the t -th derivative of function f at the root λ_i (zero derivative of a function is understood as the function itself). The MV polynomials¹⁰ $Q_i^{m_i-1-t}(\mathbf{A}, \lambda_i) = Q_i^{m_i-1-t}(\mathbf{A}, \lambda_i)|_{\lambda=\lambda_i}$ are recursively computed as follows:

$$\begin{aligned}
 Q_i^{m_i-1}(\mathbf{A}, \lambda) &= \frac{S^{(0)}(\mathbf{A}, \lambda)}{\mu^{(m_i)}(\lambda)} \\
 Q_i^{m_i-2}(\mathbf{A}, \lambda) &= \frac{S^{(1)}(\mathbf{A}, \lambda) - Q_i^{m_i-1}(\mathbf{A}, \lambda)\mu^{(m_i+1)}(\lambda)}{\mu^{(m_i)}(\lambda)} \\
 Q_i^{m_i-3}(\mathbf{A}, \lambda) &= \frac{S^{(2)}(\mathbf{A}, \lambda) - Q_i^{m_i-2}(\mathbf{A}, \lambda)\mu^{(m_i+1)}(\lambda) - Q_i^{m_i-1}(\mathbf{A}, \lambda)\mu^{(m_i+2)}(\lambda)}{\mu^{(m_i)}(\lambda)} \\
 &\vdots \\
 Q_i^0(\mathbf{A}, \lambda) &= \frac{S^{(m_i-1)}(\mathbf{A}, \lambda) - Q_i^1(\mathbf{A}, \lambda)\mu^{(m_i+1)}(\lambda) - Q_i^2(\mathbf{A}, \lambda)\mu^{(m_i+2)}(\lambda) - \dots - Q_i^{m_i-1}(\mathbf{A}, \lambda)\mu^{(2m_i-1)}(\lambda)}{\mu^{(m_i)}(\lambda)}.
 \end{aligned} \tag{18}$$

The quantities $\mu^{(k)}(\lambda)$ are weighted k -th derivatives of minimal polynomial $\mu^{(k)}(\lambda) = \frac{1}{k!} \frac{d^k \mu(\lambda)}{d\lambda^k}$ (defined in Section 3). The polynomials $S(\mathbf{A}, \lambda) = S^{(0)}(\mathbf{A}, \lambda)$ and their weighted k -th derivatives $S^{(k)}(\mathbf{A}, \lambda)$ are defined by

$$\begin{aligned}
 S(\mathbf{A}, \lambda) &= S^{(0)}(\mathbf{A}, \lambda) = \sum_{k=0}^{r-1} \left(\sum_{s=0}^{r-k-1} \lambda^s C_{(r-s-k-1)}(\mathbf{A}) \right) \mathbf{A}^k \\
 S^{(0)}(\mathbf{A}, \lambda_i) &= S^{(0)}(\mathbf{A}, \lambda)|_{\lambda=\lambda_i} \\
 S^{(k)}(\mathbf{A}, \lambda) &= \frac{1}{k!} \frac{d^k S(\mathbf{A}, \lambda)}{d\lambda^k}, \quad S^{(k)}(\mathbf{A}, \lambda_i) = S^{(k)}(\mathbf{A}, \lambda)|_{\lambda=\lambda_i}.
 \end{aligned} \tag{19}$$

Proof. At the moment, we have proved formulas in the Proposition 1 for Clifford algebras with $p+q=4$ and $p+q=6$ by symbolically verifying them for defective MVs with all possible combinations of nontrivial Jordan blocks of algebras that have real matrix representations.¹¹ Also, we have checked formulas numerically (using exact arithmetic) for $p+q \leq 12$, for this purpose using MVs with integer coefficients. Since the proposition is formulated in terms of polynomials with unspecified coefficients, a general proof could be tedious and will require manipulation/summation of products of coefficients of these polynomials. This task is postponed for further research. $\square$

Remark 1. (Generalization). It is straightforward to generalize the Proposition 1 for arbitrary (finite dimensional) square matrices and linear operators. Indeed, in order to compute a function of matrix, it is enough to replace MV geometric product by matrix product $\mathbf{A}^k \rightarrow \underbrace{\hat{\mathbf{A}} \cdot \hat{\mathbf{A}} \cdots \hat{\mathbf{A}}}_{k \text{ terms}}$. Computation of minimal polynomial of finite matrices is a well-known procedure.

Remark 2. (Recursion formula analogy). The recursion formulas (18) reminds one of the procedure of construction of basis vectors for some irreducible representation [21] of Lie group. Since we already know from Section 4 that $Q^{m_i} = 0$, the construction starts from a single basis vector of the highest weight Q^{m_i-1} . Then one defines a lowering operator, application of which in succession produces basis vectors by one lower weight in each step. The procedure is repeated until the lowest weight basis vector of the representation is reached.¹² One could observe that the polynomial $Q_i^{m_i-1}$ corresponds to the polynomial $q_i^{m_i-1}$ of the highest power in a classical method. Indeed (see Section 4, paragraph above formula (10)), multiplication of $q_i^{m_i-1}$ by q_i results in zero, $q_i q_i^{m_i-1} = q_i^{m_i} = 0$. Therefore, $Q_i^{m_i-1}$ is exactly the poly-

nomial from which the entire recursive lowering procedure starts. After subtraction of all previous Q_i^k contributions in the last step, we are left with the polynomial $Q_i^0(\mathbf{A}, \lambda)$, which has been denoted by p_i in the classical method. Since explicit expressions for $Q_i^k(\mathbf{A}, \lambda)$ at $k > 0$ coincide with those at $k \leq 0$, they have exactly the same properties as listed in Section 4, and of course, the properties of $Q_i^0(\mathbf{A}, \lambda)$ are identical to p_i . Therefore, the recursion formulas (18), in fact, implement partial fraction decomposition of inverse minimal polynomial in a reverse order, what makes polynomial reduction redundant.

Remark 3. (Denominators). One should observe that denominators of (18) enclose the weighted m_i -th derivative in the minimal polynomial $\mu^{(m_i)}(\lambda)$. In fact, it prevents the denominator at root λ_i of multiplicity m_i to vanish. Indeed, at the root λ_i of multiplicity m_i , we have $\mu_{\mathbf{A}}(x) = (x - \lambda_i)^{m_i} p(x)$, where polynomial $p(x)$ encompasses contribution from all other roots $\lambda_k \neq \lambda_i$. After differentiation m_i times, we get an expression in which $(x - \lambda_i)$ disappears. Since $p(\lambda_i) \neq 0$, the denominator does not reduce to zero at $x = \lambda_i$. This explains why in case of a non repeating roots in equation (16) in paper [5], the expression for β factor in the denominator has only the first derivative of minimal polynomial.

Remark 4. (Number of derivatives of MV function). From Equation (17), one can see that the non-diagonalizable MV function demands computation of the function derivatives of order that is by one less than the maximal multiplicity of a set of roots. From this follows that MV function that has no repeated roots, that is, the MV is non-defective and does not require computation of function derivatives at all. This is in agreement with formula (16) in paper [5].

Remark 5. (Evaluation). Symbolic recursion formulas are same¹³ for every root λ_i of the same multiplicity, and consequently, the expressions for them are identical up until a particular root value will be inserted. Therefore, it is more efficient at first to do all computations with a formal root λ symbolically and then later to substitute a particular root values in a final step of computation. Moreover, we can compute entire generalized spectral basis using dummy commutative variable x and make replacements $x^k \rightarrow \mathbf{A}^k$ in a very last step of computations (in a matrix case, the replacement reads $x^k \rightarrow \underbrace{\hat{\mathbf{A}} \cdot \hat{\mathbf{A}} \cdots \hat{\mathbf{A}}}_{k \text{ terms}}$).

Remark 6. (Characteristic polynomial). We found it somehow surprising that instead of the minimal polynomial $\mu(\mathbf{A})$ in

(18), we can use the characteristic polynomial $\chi(\mathbf{A})$ or any other polynomial $p(\mathbf{A})$, which yields zero for a given MV $\mathbf{A}$. Indeed, if $S^{(0)}(\mathbf{A}, \lambda_i)$ is zero, then $Q_i^{m_i-1}(\mathbf{A}, \lambda_i)$ is zero as well. It means that $Q_i^{m_i-2}(\mathbf{A}, \lambda_i)$ becomes the polynomial $Q_i^{m_i-1}(\mathbf{A}, \lambda_i)$, with shifted index $i' = i - 1$ (since a differentiation $S^{(1)}(\mathbf{A}, \lambda_i)$ reduces polynomial degree by one). Therefore, an arbitrary polynomial what satisfy $p(\mathbf{A}) = 0$ can be used. Of course, computations then become less effective, since, in general, we need to compute higher powers of MV for substitution in a final expression. It also suggest that, in principle, only eigenvalues and estimation of largest multiplicity are required to start the whole computation.

Since the roots of a characteristic equation in real GA in general are complex numbers, the individual terms in the sums in Proposition 1, strictly speaking, are complex. However, the coefficients at basis elements in the final result for some functions (for example, for exponential function) will simplify to real numbers. The real expressions for coefficients (when possible) can be obtained by summing complex conjugate roots pairwise to real expressions exactly in the same way as it was explained in [5].

6 | Example: Defective MV in $Cl_{3,0}$

The section demonstrates how to apply Proposition 1 to compute arbitrary GA function of a non-diagonalizable MV, that is, of MV minimal polynomial of which has at least one repeated root. In the examples below, we will compute the exponential function of MV, since the obtained answer then can be easily checked by using the defining property of the exponent, namely,

$$\mathbf{A} \exp(\mathbf{A}) = \left. \frac{\partial \exp(\mathbf{A}t)}{\partial t} \right|_{t=1}. \quad (20)$$

To find arbitrary function of MV, it is enough to replace the exponent and its scalar derivatives by corresponding function and its derivatives in the examples presented below. The only restriction is that the function (and the required derivatives) should be well defined at the roots of MV minimal polynomial.

Example 1. *Exponent in $Cl_{3,0}$.* Let us take MV $\mathbf{A} = -1 + 2\mathbf{e}_1 - 2\mathbf{e}_{12} - \mathbf{e}_{123} + \mathbf{e}_2 + \mathbf{e}_{23} + 2\mathbf{e}_3$ the exponential of which typifies non-diagonalizable MV. The minimal polynomial of $\mathbf{A}$ is $\mu_{\mathbf{A}}(x) = (x^2 + 2x + 2)^2 = x^4 + 4x^3 + 8x^2 + 8x + 4$ with $C_{(4)} = 4, C_{(3)} = 8, C_{(2)} = 8, C_{(1)} = 4, C_{(0)} = 1$. It has two complex ($r \in \{1, 2\}$) roots, namely, $\{x_1 = \lambda_1 = -1 - i, x_2 = \lambda_2 = -1 + i\}$. Each root has multiplicity two: $m_1 = 2$, and $m_2 = 2$. To avoid confusion, in the formulas below, we have replaced λ by x . First, we will compute polynomials p_1, q_1, p_2, q_2 using a classical method described in Section 4. In this particular case, one finds two pairs of poly-

nomials $\left\{ \underbrace{p_1, q_1}_{m_1}, \underbrace{p_2, q_2}_{m_2} \right\}$ of multiplicity $m_1 = m_2 = 2$.

Partial fraction decomposition of the first ($r = 1$) root, $\lambda_1 = -1 - i$, yields $p_1 = \frac{1}{4}i(x + (1 - i))^2(x + (1 + 2i))$ and $q_1 = -\frac{1}{4}(x + (1 - i))^2(x + (1 + i))$. For the second ($r = 2$) root λ_2 , the polynomial is $p_2 = -\frac{1}{4}i(x + (1 + i))^2(x + (1 - 2i))$, and $q_2 = -\frac{1}{4}(x + (1 - i))(x + (1 + i))^2$. Here, we will only check that the polynomials indeed have properties listed in Section 4. For

computational details of p_1, q_1, p_2, q_2 , the reader should refer to paper [8]. It is a straightforward matter to check that $p_1 + p_2 = 1$. The verification of multiplicative properties requires division of polynomials modulo minimal polynomial. For example, to verify that $p_1 q_1 = q_1$, we have to divide the product $p_1 q_1$ by minimal polynomial $\mu_{\mathbf{A}}(x)$, that is, find $s(x)$ and $r(x)$ that allow to write the product in the form $p_1(x)q_1(x) = s(x)\mu_{\mathbf{A}}(x) + r(x)$. Here, $s(x)$ is the polynomial of lower degree, which after multiplication by minimal polynomial $\mu(x)$ and summation with the remainder $r(x)$ yields the product $p_1(x)q_1(x)$. Division by modulo minimal polynomial $\mu(x)$, denoted $(p_1(x)q_1(x) \bmod \mu(x)) = r(x)$, means that we are only interested in (keep) the remainder $r(x)$ and discard $s(x)\mu(x)$ part. In particular, to check the property $q_1^2 = 0$, we have to verify that $(q_1(x)^2 \bmod \mu(x)) = 0$, that is, that $\mu(x)$ divides $q_1(x)^2$ (the remainder is zero). This is easy to do using computer algebra programs. For example in *Mathematica*, this can be done with command **PolynomialReduce**[$p_1(x) q_1(x), \mu(x), x$]. Then it is easy to check that $(p_1(x)q_1(x) \bmod \mu(x)) = q_1(x)$ and $(p_1(x)p_2(x) \bmod \mu(x)) = 0$. The calculations are similar for polynomials $p_2(x), q_2(x)$ of the second root λ_2 .

Once the generalized spectral basis was found, it is easy to check that the powers of MV satisfy the identities:

$$\begin{aligned} \mathbf{A} &= (\lambda_1 + q_1(\mathbf{A}))p_1(\mathbf{A}) + (\lambda_2 + q_2(\mathbf{A}))p_2(\mathbf{A}) \\ \mathbf{A}^2 &= (\lambda_1 + q_1(\mathbf{A}))^2 p_1(\mathbf{A}) + (\lambda_2 + q_2(\mathbf{A}))^2 p_2(\mathbf{A}) \\ &\vdots \end{aligned}$$

where commutative variable x was replaced by multivector $\mathbf{A}$. The same replacement, which lies in the core of the Cayley–Hamilton theorem, was considered in Section 2.

Expressions for MV powers permit straightforward calculation of MV function by formula (15); for example, to compute the exponential $\exp(\mathbf{A})$, we just need to replace the function g by $\exp$. In the case of defective MV, the formula (15) requires computation of derivatives. Since, in our case, the roots have multiplicity $m_{1,2} = 2$, only value of the first scalar derivative of the $\exp$ function is required at roots $\lambda_{1,2}$. It is convenient at first to calculate all needed derivatives symbolically for arbitrary λ and then substitute particular root λ_i . For $\exp(\lambda)$, the derivative is $\exp'(\lambda_i) = \frac{1}{1!} \frac{d \exp(\lambda + \tau)}{d\tau} \Big|_{\tau=0, \lambda=\lambda_i} = \exp(\lambda_i)$. Explicit formula in this case is

$$\begin{aligned} \exp(\mathbf{A}) &= (\exp(\lambda_1)p_1(\mathbf{A}) + \exp'(\lambda_1)q_1(\mathbf{A})) \\ &\quad + (\exp(\lambda_2)p_2(\mathbf{A}) + \exp'(\lambda_2)q_2(\mathbf{A})) \end{aligned}$$

After substitution of $\mathbf{A} = -1 + 2\mathbf{e}_1 - 2\mathbf{e}_{12} - \mathbf{e}_{123} - 2\mathbf{e}_{13} + \mathbf{e}_2 + \mathbf{e}_{23} + 2\mathbf{e}_3$, one obtains

$$\begin{aligned} \exp(\mathbf{A}) &= + \frac{1}{2}(1 + e^{2i})e^{-1-i} + \frac{1}{2}((2 + i) + (2 - i)e^{2i})e^{-1-i}\mathbf{e}_1 \\ &\quad + \frac{1}{2}((1 + 2i) + (1 - 2i)e^{2i})e^{-1-i}\mathbf{e}_2 + (1 + i)(e^{2i} - i)e^{-1-i}\mathbf{e}_3 \\ &\quad - (1 - i)(i + e^{2i})e^{-1-i}\mathbf{e}_{12} - \frac{1}{2}((2 - i) + (2 + i)e^{2i})e^{-1-i}\mathbf{e}_{13} \\ &\quad + \frac{1}{2}((1 - 2i) + (1 + 2i)e^{2i})e^{-1-i}\mathbf{e}_{23} \\ &\quad + \frac{1}{2}i(-1 + e^{2i})e^{-1-i}\mathbf{e}_{123}. \end{aligned}$$

After summing complex conjugate roots pairwise, the answer can be rewritten in a real form:

$$\begin{aligned} \exp(A) = & \frac{1}{e} (\cos(1) + \mathbf{e}_1(\sin(1) + 2\cos(1)) + \mathbf{e}_2(2\sin(1) + \cos(1)) \\ & + 2\mathbf{e}_3(\cos(1) - \sin(1)) \\ & - 2\mathbf{e}_{12}(\sin(1) + \cos(1)) - \sin(1)\mathbf{e}_{123} \\ & + \mathbf{e}_{13}(\sin(1) - 2\cos(1)) + \mathbf{e}_{23}(\cos(1) - 2\sin(1))). \end{aligned}$$

The essential part of the above computation (Section 4) is the partial fraction decomposition of $1/\mu(A)$, which yields the polynomials $p_i = Q_i^0, q_i^1 = Q_i^1, q_i^2 = Q_i^2, \dots, q_i^{m_i-1} = Q_i^{m_i-1}$ and which usually is a nontrivial task. Even so, we shall demonstrate how easily the required polynomials can be obtained by recursion procedure (18) from the main Proposition 1.

The first task is to calculate the largest polynomial power $Q_i^{m_i-1}(A, \lambda)$ for each root (the highest weight vector) that correspond to $q_i^{m_i-1}$ from classical method description in Section 4. Since the roots have the multiplicity 2, the first line of the system (18) yields

$$\begin{aligned} Q_{i=1,2}^1(x, \lambda) &= \frac{S^{(0)}(x, \lambda)}{\frac{1}{2!}\mu''(\lambda)} \\ &= \frac{C_{(0)}x^3 + x^2(C_{(0)}\lambda + C_{(1)}) + x(C_{(0)}\lambda^2 + C_{(1)}\lambda + C_{(2)}) + C_{(0)}\lambda^3 + C_{(1)}\lambda^2 + C_{(2)}\lambda + C_{(3)}}{4(\lambda^2 + 2\lambda + 2) + 2(2\lambda + 2)^2} \\ &= \frac{x^3 + (\lambda + 4)x^2 + (\lambda^2 + 4\lambda + 8)x + \lambda^3 + 4\lambda^2 + 8\lambda + 8}{4(\lambda^2 + 2\lambda + 2) + 2(2\lambda + 2)^2}, \\ Q_1^1(x, -1 - i) &= -\frac{1}{4}(x + (1 - i))^2(x + (1 + i)), \\ Q_2^1(x, -1 + i) &= -\frac{1}{4}(x + (1 - i))(x + (1 + i))^2. \end{aligned}$$

The polynomials $Q_i^0(x, \lambda)$, which represent p_i in the classical method, can be found from the second line of the system (18).

$$\begin{aligned} Q_{i=1,2}^0(x, \lambda) &= \frac{\frac{1}{1!}\frac{dS^{(0)}(x, \lambda)}{d\lambda} - Q_{i=1,2}^1(x, \lambda)\frac{1}{3!}\frac{d^3\mu(\lambda)}{d\lambda^3}}{\frac{1}{2!}\frac{d^2\mu(\lambda)}{d\lambda^2}} \\ &= \frac{(-2\lambda - 2)x^3 + (\lambda^2 - 4\lambda - 4)x^2 + (4\lambda^3 + 14\lambda^2 + 8\lambda)x + 7\lambda^4 + 32\lambda^3 + 60\lambda^2 + 48\lambda + 16}{2(3\lambda^2 + 6\lambda + 4)^2}, \\ Q_1^0(x, -1 - i) &= \frac{1}{4}i(x + (1 - i))^2(x + (1 + 2i)), \\ Q_2^0(x, -1 + i) &= -\frac{1}{4}i(x + (1 + i))^2(x + (1 - 2i)). \end{aligned}$$

A bonus is that we have avoided polynomial division modulus minimal polynomial, since the recursive computation starts from the “opposite end” as compared to classical method. Therefore, here proposed method is much simpler than that presented in detail in paper [8], which in turn may be advantageous over other known methods.

7 | Example: Defective MV in $Cl_{4,2}$

To show how well the recursive formulas work, we will take a rather complicated MV, which corresponds to a real matrix of dimension 8×8 and compute exponential function of the MV.

Example 2. *Exponential in $Cl_{4,2}$.* Let us take rather complicated and non-diagonalizable MV $A = \frac{1}{8}(2\mathbf{e}_1 - \mathbf{e}_{13} - \mathbf{e}_{134} + 2\mathbf{e}_{1345} - 10\mathbf{e}_{13456} + 4\mathbf{e}_{135} + 2\mathbf{e}_{136} - 4\mathbf{e}_{14} + \mathbf{e}_{145} - 2\mathbf{e}_{1456} + 2\mathbf{e}_{146} - \mathbf{e}_{15} + 4\mathbf{e}_{16} - 2\mathbf{e}_{34} - 4\mathbf{e}_{345} + 2\mathbf{e}_{3456} - \mathbf{e}_{346} - 2\mathbf{e}_{35} - 4\mathbf{e}_{356} + \mathbf{e}_{36} + \mathbf{e}_{456} + 2\mathbf{e}_5 + \mathbf{e}_{56} + 2\mathbf{e}_6 + 30)$ the minimal polynomial of which is $\mu_A(x) = (x - 5)^4(x - 3)^3(x - 1) = x^8 - 30x^7 + 386x^6 - 2774x^5 + 12132x^4 - 32890x^3 + 53550x^2 - 47250x + 16875$ with $C_{(8)} = 16875, C_{(7)} = -47250, C_{(6)} = 53550, C_{(5)} = -32890, C_{(4)} = 12132, C_{(3)} = -2774, C_{(2)} = 386, C_{(1)} = -30,$ and $C_{(0)} = 1$.

The MV has three real ($r \in \{1, 2, 3\}$) roots, namely, $\{x_1 = \lambda_1 = 1, x_2 = \lambda_2 = 3, x_3 = \lambda_3 = 5\}$. Corresponding multiplicities are $m_1 = 1, m_2 = 3,$ and $m_3 = 4$. We need to compute polynomials:

$$\left\{ \begin{array}{ccc} Q_1^0, & Q_2^0, Q_2^1, Q_2^2, & Q_3^0, Q_3^1, Q_3^2, Q_3^3 \\ m_1 & m_2 & m_3 \end{array} \right\}.$$

First, we compute all parts that might enter into recursive formulas up to maximal multiplicity $m_i = 4$. The weighted derivatives of minimal polynomial are

$$\begin{aligned} \mu^{(1)}(\lambda) &= 8\lambda^7 - 210\lambda^6 + 2316\lambda^5 - 13870\lambda^4 + 48528\lambda^3 \\ &\quad - 98670\lambda^2 + 107100\lambda - 47250 \\ \mu^{(2)}(\lambda) &= 28\lambda^6 - 630\lambda^5 + 5790\lambda^4 - 27740\lambda^3 + 72792\lambda^2 \\ &\quad - 98670\lambda + 53550 \end{aligned}$$

$$\begin{aligned}\mu^{(3)}(\lambda) &= 56\lambda^5 - 1050\lambda^4 + 7720\lambda^3 - 27740\lambda^2 + 48528\lambda - 32890 \\ \mu^{(4)}(\lambda) &= 70\lambda^4 - 1050\lambda^3 + 5790\lambda^2 - 13870\lambda + 12132, \\ \mu^{(5)}(\lambda) &= 56\lambda^3 - 630\lambda^2 + 2316\lambda - 2774 \\ \mu^{(6)}(\lambda) &= 28\lambda^2 - 210\lambda + 386 \\ \mu^{(7)}(\lambda) &= 8\lambda - 30.\end{aligned}$$

(21)

Polynomials $S^{(k)}(A, \lambda)$ are¹⁴

$$\begin{aligned}S^{(0)}(A, \lambda) &= A^7 - 30A^6 + 386A^5 - 2774A^4 + 12132A^3 \\ &+ (A^2 - 30A + 386)\lambda^5 - 32890A^2 \\ &+ (A^3 - 30A^2 + 386A - 2774)\lambda^4 \\ &+ (A^4 - 30A^3 + 386A^2 - 2774A + 12132)\lambda^3 \\ &+ (A^5 - 30A^4 + 386A^3 - 2774A^2 + 12132A - 32890)\lambda^2 \\ &+ (A^6 - 30A^5 + 386A^4 - 2774A^3 + 12132A^2 - 32890A + 53550) \\ &\lambda + (A - 30)\lambda^6 + 53550A + \lambda^7 - 47250 \\ S^{(1)}(A, \lambda) &= A^6 - 30A^5 + 386A^4 - 2774A^3 + 5(A^2 - 30A + 386) \\ &\lambda^4 + 12132A^2 + 4(A^3 - 30A^2 + 386A - 2774)\lambda^3 \\ &+ 3(A^4 - 30A^3 + 386A^2 - 2774A + 12132)\lambda^2 \\ &+ 2(A^5 - 30A^4 + 386A^3 - 2774A^2 + 12132A - 32890)\lambda \\ &+ 6(A - 30)\lambda^5 - 32890A + 7\lambda^6 + 53550 \\ S^{(2)}(A, \lambda) &= A^5 - 30A^4 + 386A^3 + 10(A^2 - 30A + 386)\lambda^3 \\ &- 2774A^2 + 6(A^3 - 30A^2 + 386A - 2774)\lambda^2 \\ &+ 3(A^4 - 30A^3 + 386A^2 - 2774A + 12132)\lambda \\ &+ 15(A - 30)\lambda^4 + 12132A + 21\lambda^5 - 32890 \\ S^{(3)}(A, \lambda) &= A^4 - 30A^3 + 10(A^2 - 30A + 386)\lambda^2 + 386A^2 \\ &+ 4(A^3 - 30A^2 + 386A - 2774)\lambda + 20(A - 30)\lambda^3 \\ &- 2774A + 35\lambda^4 + 12132.\end{aligned}$$

Now, for each root, let us compute $Q_i^k(A, \lambda)$ polynomials. Start from the root $\lambda = 5$, the number of which is $i = 3$, and the multiplicity is $m_3 = 4$ (the largest). One has to compute four polynomials:

$$\begin{aligned}Q_3^3(A, 5) &= \frac{1}{32}(A - 5)^3(A - 3)^3(A - 1) \\ Q_3^2(A, 5) &= -\frac{1}{128}(A - 5)^2(A - 3)^3(A - 1)(7A - 39) \\ Q_3^1(A, 5) &= \frac{1}{512}(A - 5)(A - 3)^3(A - 1)(31A^2 - 338A + 931) \\ Q_3^0(A, 5) &= -\frac{1}{2048}(A - 3)^3(A - 1) \\ &\quad (111A^3 - 1789A^2 + 9677A - 17599).\end{aligned}$$

The same polynomials for root $\lambda = 3$, which number is $i = 2$ and the multiplicity $m_2 = 3$, are

$$\begin{aligned}Q_2^2(A, 3) &= \frac{1}{32}(A - 5)^4(A - 3)^2(A - 1) \\ Q_2^1(A, 3) &= \frac{1}{64}(A - 5)^4(A - 3)(A - 1)(3A - 7) \\ Q_2^0(A, 3) &= \frac{1}{128}(A - 5)^4(A - 1)(7A^2 - 36A + 49).\end{aligned}$$

For the first $i = 1$ root $\lambda = 1$ with multiplicity $m_1 = 1$, we need only

$$Q_1^0(A, 1) = -\frac{(A - 5)^4(A - 3)^3}{2048} \quad (22)$$

This ends up computation of generalized spectral basis. Before going over to computation of exponential, let us check the identity $A = 1 Q_1^0(A, 1) + (3 + Q_2^1(A, 3))Q_2^0(A, 3) + (5 + Q_3^1(A, 5))Q_3^0(A, 5)$. Substitution of all Q_j^k into RHS yields

$$\begin{aligned}&\frac{1}{1048576}(-753A^{14} + 37946A^{13} - 874979A^{12} + 12226436A^{11} \\ &- 115570953A^{10} + 781115526A^9 - 3889496339A^8 \\ &+ 14481221176A^7 - 40461630635A^6 + 84287184806A^5 \\ &- 128533325601A^4 + 138586998020A^3 - 99271453275A^2 \\ &+ 41924292826A - 7799675625)\end{aligned} \quad (23)$$

Substituting initial MV A into latter expression and expanding all geometric powers, one indeed obtains A , which is the LHS of the identity.

Similar check of identity $A^2 = 1^2 Q_1^0(A, 1) + (3 + Q_2^1(A, 3))^2 Q_2^0(A, 3) + (5 + Q_3^1(A, 5))^2 Q_3^0(A, 5)$ yields true answer too. This indicates that generalized spectral basis was computed correctly, and we can proceed with computation of MV function using formula (16). We only need to compute some derivatives of exponential function. Then, resorting to already described procedure, we get final answer for MV exponential function,

$$\begin{aligned}\exp(A) &= -\frac{1}{48}e((6 - 6e^2)\mathbf{e}_1 + 6(e^2 + e^4)\mathbf{e}_{13} \\ &+ 6(1 + e^2 - 2e^4)\mathbf{e}_{1345} + 6(-1 - 3e^2 + 4e^4)\mathbf{e}_{13456} \\ &+ e^2(7e^2 - 6)\mathbf{e}_{1346} - 6(-1 + e^2 + 2e^4)\mathbf{e}_{135} \\ &+ 6(1 + e^2)\mathbf{e}_{1356} - 5e^4\mathbf{e}_{136} + (18e^2 - 6)\mathbf{e}_{14} \\ &+ e^2(6 + 7e^2)\mathbf{e}_{1456} + (6 - 6e^2)\mathbf{e}_{146} \\ &- 6e^2(e^2 - 1)\mathbf{e}_{15} + 5e^4\mathbf{e}_{156} + (6 - 18e^2)\mathbf{e}_{16} \\ &+ e^2(7e^2 - 6)\mathbf{e}_3 + 5e^4\mathbf{e}_{34} + (18e^2 - 6)\mathbf{e}_{345} \\ &+ (6 - 6e^2)\mathbf{e}_{3456} + 6(e^2 + e^4)\mathbf{e}_{346} + 6(e^2 - 1)\mathbf{e}_{35} \\ &+ (18e^2 - 6)\mathbf{e}_{356} + 6(1 + e^2)\mathbf{e}_4 + 5e^4\mathbf{e}_{45} \\ &+ 6e^2(e^2 - 1)\mathbf{e}_{456} + 6(-1 + e^2 - 2e^4)\mathbf{e}_{46} \\ &- e^2(6 + 7e^2)\mathbf{e}_5 - 6(1 + e^2 + 2e^4)\mathbf{e}_6 \\ &- 6(1 + 3e^2 + 4e^4)).\end{aligned}$$

The answer was checked that it satisfies a defining property of MV exponential, Equation (20).

8 | Example: Defective MV in $Cl_{4,2}$ With High Degree Irreducible Polynomial

The last example demonstrates that for some inputs, even sophisticated computer algebra systems can have problems computing

functions of matrices, where our method handles the cases efficiently and flawlessly. Unfortunately, the computed answers are too large to be presented explicitly; therefore, we will compare results by evaluating timing and complexity of answers by counting number of leafs of returned symbolic expressions.

Example 3. To illustrate the point, in $CI_{4,2}$ algebra, we take a non-diagonalizable MV,

$$\begin{aligned} A = & -1 - \mathbf{e}_3 + \mathbf{e}_6 - \mathbf{e}_{12} - \mathbf{e}_{13} + \mathbf{e}_{15} - \mathbf{e}_{24} - \mathbf{e}_{25} \\ & + \mathbf{e}_{26} - \mathbf{e}_{34} - \mathbf{e}_{35} + \mathbf{e}_{36} - \mathbf{e}_{45} + \mathbf{e}_{56} \\ & + \mathbf{e}_{123} + \mathbf{e}_{124} + \mathbf{e}_{126} + \mathbf{e}_{134} \\ & + \mathbf{e}_{135} + \mathbf{e}_{136} + \mathbf{e}_{146} + \mathbf{e}_{234} - \mathbf{e}_{235} - \mathbf{e}_{236} - \mathbf{e}_{245} - \mathbf{e}_{246} \quad (24) \\ & - \mathbf{e}_{256} + \mathbf{e}_{456} - \mathbf{e}_{1236} + \mathbf{e}_{1245} - \mathbf{e}_{1246} + \mathbf{e}_{1256} - \mathbf{e}_{1345} \\ & - \mathbf{e}_{1346} - \mathbf{e}_{1356} + \mathbf{e}_{1456} - \mathbf{e}_{2346} - \mathbf{e}_{2356} + \mathbf{e}_{2456} \\ & + \mathbf{e}_{3456} + \mathbf{e}_{12345} - \mathbf{e}_{12346} + \mathbf{e}_{12356} \end{aligned}$$

and compute $\exp(A)$.

The MV has the following 8×8 real matrix representation:

$$\begin{pmatrix} 0 & 0 & 0 & -2 & -2 & 0 & 2 & -1 \\ 0 & -2 & 4 & 2 & -2 & 2 & 5 & 2 \\ 4 & 0 & 0 & 2 & 4 & -3 & -2 & 6 \\ -2 & -2 & 2 & -2 & -1 & -4 & 0 & 2 \\ 0 & -2 & 2 & -1 & 0 & 2 & -2 & -2 \\ 0 & -4 & 1 & 2 & -2 & 2 & -4 & 0 \\ 0 & 1 & 0 & 0 & 2 & 0 & -4 & 0 \\ -1 & 0 & -2 & 4 & -2 & 4 & -4 & -2 \end{pmatrix} \quad (25)$$

in $CI_{4,2}$ algebra basis that is expressed by matrices

$$\begin{aligned} \hat{\mathbf{e}}_1 = & \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \end{pmatrix} \\ \hat{\mathbf{e}}_2 = & \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} \hat{\mathbf{e}}_3 = & \begin{pmatrix} 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{pmatrix} \\ \hat{\mathbf{e}}_4 = & \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \end{pmatrix} \quad (26) \\ \hat{\mathbf{e}}_5 = & \begin{pmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \end{pmatrix} \\ \hat{\mathbf{e}}_6 = & \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \end{pmatrix} \end{aligned}$$

The respective MV/matrix has minimal polynomial $(\lambda - 1)^2(\lambda^6 + 10\lambda^5 + 39\lambda^4 + 124\lambda^3 + 543\lambda^2 - 198\lambda - 4743)$. While *Mathematica* version 13.0 in our previous experiments [5] has crashed on this input after 48 h of computation after it has exhausted all 96GB of server RAM, the version 14.1 successfully completes the same task on laptop with 16GB RAM in about 64 s. This is to be compared to 1.5 s for the same calculations using our method for MV or just to 0.5 s when using matrices (the difference is due to highly optimized multiplication of matrices in *Mathematica*) on the same hardware. The leaf count (measure of complexity) of the result also differs by orders of magnitude. Leaf count of our complex valued result (answer in real form is twice smaller) is less than 10^6 (when converted to matrix form for comparison), while leaf count of *Mathematica* function **MatrixFunction[]** output¹⁵ is almost 2.5×10^9 . Even numerical comparison of both matrices up to 100 number digit precision took over 780 s (numerical evaluation time of our output to this precision is negligible, 1/10 s).

9 | Conclusions and Perspective

The paper provides a recursive method to compute generalized spectral basis of multivectors (MVs) in Clifford geometric algebra. The method can be easily extended to the square defective (non-diagonalizable) matrices as well. The generalized spectral basis opens fast and easy way to compute functions of the MVs, matrices and linear operators. The main result of this paper is presented in Proposition 1 (formulas (17), (18), and (19)). Using defective MVs, a number of examples with real MVs are presented that demonstrate how the method works in practice. Also, comparison with the classical method is provided. As far as we know, the proposed recursive method provides *exact*, simplest, and fastest tool to compute functions of MV and matrix as well, since computation of the generalized spectral basis does not require polynomial reduction. At present, approximate numerical methods are mainly used to evaluate functions of matrix [4]. However, if exact results are need, even the sophisticated computer algebra systems have problems in computing exact MV/matrix function of quite simple form, when minimal polynomial includes a high degree (irreducible) polynomial as was demonstrated in Section 8.

Though at the moment we are unable to provide a strict proof of Proposition 1 for the most general case, we have no doubts that our presented recursive method provides better and more preferable way to compute generalized spectral basis and, in its turn, gives the go-ahead to do calculations with functions of MV/matrix argument. In fact, the definition of minimal polynomial of MV and existence of new recursive sequence itself provides an exciting research and application perspective. For example, recently, there were attempts [22, 23] to define MV rank by singular value decomposition of MV without any reference to matrix representations. This work suggests that the rank of MV can be defined simply as a degree of minimal polynomial, $\text{rank}(A) = \deg(\mu(A)) = \sum_{i=0}^r m_i$ (or may be as a difference between highest and smallest exponents of monomials) and computed by Algorithm 1, without any reference to matrix representation as well.

To summarize, we have employed a number of examples using exact numerical MVs/matrices to demonstrate how the method works. The recursive generalized spectral basis computation procedure can be applied to matrices of general linear group/algebra, that is, to matrices (with symbolic entries as well) without any restrictions or conditions on its elements, and there are plenty problems in physics that require to find functions of matrices, for which close form solutions would be interesting to know. We hope that the generalized spectral basis will help to solve some of the mentions problems.

Author Contributions

A. Acus: conceptualization, investigation, writing – original draft, software. **A. Dargys:** validation, writing – review and editing.

Conflicts of Interest

The authors declare no conflicts of interest.

Endnotes

¹ This choice ensures that the (symbolic) coefficient $C_{(d)}(A)$ (determinant of MV) remains syntactic positive in any vector space of dimension n .

² Multivector representation by matrices is determined by Bott table [2], p. 205, which defines isomorphism (understood as isomorphism among vector spaces, not algebras) between multivectors and matrices. The concepts of MV determinant and trace here are defined [17] without reference to the mentioned matrix representation.

³ An attempt to compute MV function in this way, in general, is a bad plan, since at first sight even for a well behaved and convergent function some of the coefficients may skyrocket to very high values before starting to decrease to a true value. An example of bad behavior can be found in paper [7].

⁴ We intentionally use the same notation for coefficients for both characteristic and minimal polynomials, since for diagonalizable MV, coefficients of both polynomials coincide (in the article, they have opposite signs for computational purposes), and we will deal further with the minimal polynomials only.

⁵ The algorithm in the paper [5] did not take into account situation when number of vectorList elements exceed the degree of characteristic polynomial. In the latter case, one must reduce the degree of polynomial by one. This minor correction is now represented by **If** statement in Algorithm 1.

⁶ This is guaranteed by the Cayley–Hamilton theorem that claims that powers of A^d or greater are linear combinations of lower powers of A .

⁷ All polynomial products should be understood as products modulo $\mu(x)$. For more details on how these properties can be verified, see example in Section 6.

⁸ The Cayley–Hamilton theorem, of course, ensures that in the sum there exists a maximal power of A^d . Nonetheless, in arbitrary basis, the series remains infinite because each A^k power, $k > d$, that exceed dimension d will contribute to all coefficients of lower powers of A . Now, we see the main reason why the generalized spectral basis is so useful is that the coefficient sums terminate in this basis at $k \leq d$.

⁹ See <https://reference.wolfram.com/language/tutorial/SomeNotesOnInternalImplementation.html>.

¹⁰ Since A appears in $Q_i^{m_i-1-i}(A, \lambda_i)$ explicitly, the latter is a polynomial both in λ_i and A (of course, the coefficients $C_{(k)}$ still are functions of A as well). The same notation applies to polynomials $S^{(k)}(A, \lambda)$ below.

¹¹ At the moment, we do not know an easy way to construct defective MV systematically for algebra that has complex and especially quaternion matrix representation. Nevertheless, since we know how to compute minimal polynomial for randomly generated MV, we can check the formula for arbitrary algebras, albeit not so systematically.

¹² We have chosen a case that most closely matches our algorithm. Sometimes, the irreducible representation basis is computed starting from the lowest weight vector and then applying raising operator.

¹³ Even if multiplicities of roots are distinct, the symbolic expression has a lot of repeated parts, and computations can still be highly optimized; see example in Section 7.

¹⁴ We computed all polynomials using commutative dummy variable, which in the last step was substituted by MV A to keep formulas consistent with the form given in Proposition 1.

¹⁵ Specialized function **MatrixExp[]** is slightly slower than **MatrixFunction[]** and returns the same answer.

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